Computer Networks (BCSC0008)

Assignment 3

Q-1-In the network 192.168.5.96/28, what is the fourth octet (in decimal) of the last IP address of the network that can be assigned to a host?

Q-2-A user complains that their system cannot connect to the internet, while everyone else on the same network has no issues. When you check the IP settings, you find the IP address 10.22.15.63/26 configured on the system. What would be your first step in resolving the issue?

Q-3-An IP datagram of size 1500 bytes arrives at a router. The router needs to forward this packet over a link with an MTU (Maximum Transmission Unit) of 200 bytes. Assuming the IP header is 20 bytes, how many fragments will this datagram be divided into for transmission?

Q-4-An IP router with a Maximum Transmission Unit (MTU) of 620 bytes receives an IP packet of size 2500 bytes, including a 20-byte IP header. What will be the values of the relevant fields — More Fragments (MF) bit, Fragment Offset, and Total Length — in the fourth IP fragment generated by the router?

Q-5-In an RSA cryptosystem, a particular user B uses two prime numbers p = 11 and q = 19 to generate his public and private keys. If the public key of B is 33, then the private key of B is _____.

Q-6- In an RSA cryptosystem, a particular user C uses two prime numbers p = 7 and q = 23 to generate his public and private keys. If the private key of C is 55, then the public key of C is _____.

Q-7- Suppose a network with IP Address 172.30.0.0 is divided into 4 subnets. Find the number of hosts per subnet. Also, for the first subnet, find:

1. Subnet Address
2. First Host ID
3. Last Host ID
4. Broadcast Address

Q-8- An IPv4 packet has arrived with the first 8 bits as shown: 11000001. The receiver discards the packet. Why?

Q-9- Suppose a TCP connection starts with a cwnd of 8 MSS and an ssthresh of 20 MSS. The connection experiences packet loss after the 4th RTT.

1. Calculate the cwnd after the first RTT.
2. Determine the cwnd after the second RTT.
3. Find the new ssthresh value and the cwnd after the slow start phase.

Q-10- An IPv4 packet has arrived with the first few hexadecimal digits as shown: 0x48000034000100000101…. How many hops can this packet travel before being dropped? The data belong to what upper-layer protocol?

Q-11- A small company has the IP address 192.168.0.0/24 for its main network. They need to create multiple subnets to accommodate different departments and requirements:

1. Sales Department with 150 hosts.
2. Marketing Department with 80 hosts.
3. Customer Support Department with 40 hosts.
4. R&D Department with 30 hosts.
5. Guest Wi-Fi with 60 hosts.

Q-12- A segment has arrived with the first few hexadecimal digits of its TCP header as shown:
0x04D20050AABBCCDD0011223345008018....

Based on this header:

1. What is the source port and destination port?
2. What is the sequence number?
3. How many bytes are in the TCP header?

Q-13- Explain the working of the HTTP protocol in detail. Discuss the client-server communication model used by HTTP. Describe the differences between HTTP and HTTPS.

Q-14- Discuss the working of the SMTP protocol and its role in sending emails. Describe the step-by-step process involved in sending an email from a client to a mail server and then to the recipient's mail server.

Q-15- What is the function of the POP3 protocol in email communication? Explain how POP3 retrieves messages from a mail server to a client.

Q-16- Explain the architecture and working of the File Transfer Protocol (FTP). Discuss the roles of control and data connections in FTP communication.

Q-17- Describe the purpose and structure of the MIME standard. How does MIME extend the capabilities of SMTP in sending multimedia content such as images, audio, and attachments over email?

Q-18- Explain the structure and working of the Domain Name System (DNS). Describe how a domain name is resolved to an IP address through recursive and iterative queries.

Q-19- Describe the three-way handshaking mechanism used in TCP to establish a reliable connection between two hosts. Explain the purpose of each step—SYN, SYN-ACK, and ACK—and how sequence numbers and acknowledgment numbers are used.

Q-20- Discuss how data compression improves the efficiency of data transmission over a network. Differentiate between lossless and lossy compression techniques with suitable examples.

Q-21- Discuss how encryption ensures confidentiality, integrity, and security of data during transmission. Differentiate between **symmetric** and **asymmetric** encryption techniques with examples.

Q-22- Why is the Slow Start algorithm necessary in TCP? Explain how it helps in preventing network congestion and compare it with a scenario where it is not used.

Q-23- An IP datagram is received with the following header information (in hexadecimal):

45 00 00 54 00 00 40 00 40 01 F7 C6 C0 A8 00 01 C0 A8 00 C7

Using this information, answer the following:

1. What is the version and header length of this IP packet?
2. What is the total length of the IP datagram in decimal?
3. What is the TTL (Time To Live) value?
4. What is the protocol used in this packet?
5. What are the source and destination IP addresses?

Q-24- You are given the IP address 192.168.10.0/24. You need to divide this network into **5 subnets**, each supporting at least 30 hosts.

- What subnet mask (CIDR notation) should be used?
- List the network address, first host, last host, and broadcast address of each subnet.

Q-25- Differentiate between the following connecting devices in a network:

- Hub
- Switch
- Router
- Bridge
- Gateway

Explain with respect to OSI layers, intelligence, and functionality.